

Leveraging Hybrid Algorithms for Advanced Classification of Potable Water in Smart Cities

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Leveraging Hybrid Algorithms for Advanced Classification of Potable Water in Smart Cities

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Abstract—Clean and drinkable water is essential to the health and viability of increasing urban populations. Traditional water quality analysis is time-consuming, labor-intensive, and non-real-time. This paper presents a hybrid potability classification model of water based on statistical methods and deep learning to classify potability from chemical and mineral content. With both methods, the model improves classification accuracy, decreases computational cost, and generalizes well with large data. The hybrid model created outperforms existing approaches and provides a strong, scalable solution to smart city real-time water quality monitoring.

Index Terms—Hybrid Algorithm, Water Potability Classification, Smart Cities, Deep Learning

I. INTRODUCTION

Freshwater is a minimal need and one of the foundational pillars of the sustainable development of cities. Intelligent cities are required to respond urgently to the demands for intelligent networks with real-time monitoring and classifying water quality due to the increasing population density and environmental demands. Conventional methods of analyzing water quality would often be suboptimal for addressing the needs for scalability, speed, and real-time decision-making. Also, although statistical models can produce initial observations depending on chemical and mineral conditions, they typically have difficulty detecting complex and nonlinear patterns of ¹³a.

Recent advances in machine learning, deep learning in particular, have been developed to deal with such complexity. However, deep learning models typically require gigantic amounts of labeled data and extensive computational resources, limiting their application in most problems. To close the gap, we present a hybrid classification method that integrates statistical models with deep learning paradigms to predict water samples as potable or non-potable.

Our hybrid framework not only increases the accuracy of prediction but also distributes computational requirements evenly by strategically blending light statistical methods with the deep ¹¹analysis of neural networks. The system is poised to identify complex relationships between different water quality

metrics, providing a richer and more streamlined assessment tool for environmental and public health monitoring.

This research formulates a hybrid model that combines statistical and deep learning methods to drive water potability classification based on an optimized pipeline of processing using enhanced preprocessing (i.e., KNN imputation for handling missing values) and ¹⁰re engineering. Model performance is strictly evaluated using accuracy, precision, recall, and F1-score measures and augmented with visual graphical representation to make the results easy to read. Furthermore, the scalability and deployment capability of the framework is elaborated upon, with emphasis on its suitability for smart city water monitoring systems. The integrated approach can overcome the challenges of real-world datasets while offering enhanced predictive accuracy via ensemble learning.

II. LITERATURE SURVEY

In [1], Decision Trees and Gaussian Naïve Bayes classifiers were utilized to construct a model for classifying water potability. The models were validated using K-fold cross-validation. Accuracy for the Gaussian Naïve Bayes classifier was merely 62.53 percentage, showing poor ability in identifying complex relationships between data. The research concluded that most of the Naïve Bayes variants did not perform well in water classification accurately.

A deep learning method was proposed in [2], wherein a four-hidden-layered neural network was designed to differentiate water samples. The study utilized Pearson correlation coefficient for analyzing feature dependency. Though the model was 80.07 percentage correct, it was found not to be suitable even for preliminary screening purposes. Improvement was proposed by the authors through hyperparameter tuning and architectural changes.

But another sophisticated method was investigated in [3], when a Convolutional Neural Network (CNN) model was utilized in water potability prediction. CNN performed better than other models with 97 percentage accuracy and Random

Forest with 81 percentage accuracy. The study emphasized the superior performance capacity of deep learning techniques in detecting patterns but highlighted the high-cost nature of such models.

The work in [4] used a seven-layer neural network consisting of one input layer, five hidden layers, and one output layer. The model used nine input variables and had a fairly low accuracy of 69 percentage, which was considered unsatisfactory for practical deployment. The research highlighted the necessity of more efficient optimization methods to improve performance.

In [5], a hybrid model combining Support Vector Machines (SVM) with Artificial Neural Networks (ANN) was proposed. While active learning strategies were applied to select the most effective parameters, the resulting accuracy was suboptimal. The authors noted that entropy and confidence score calculations needed further refinement to improve classification reliability.

Taken together, these studies illustrate that each individual model has unique benefits but also shows important shortcomings in accuracy, scalability, or computational performance. This lack of current research encourages the construction of a hybrid model combining statistical and deep learning methods in hopes of finding a more balanced and efficient solution for potable water classification.

III. METHODOLOGY

We propose in this paper a deep learning and machine learning combined paradigm for water potability classification as compared to conventional classifiers and more powerful neural networks. We combine Logistic Regression and Random Forest with deeper ones like feedforward neural networks (FNNs) to increase the power to predict as well as dependency.

In order to handle missing values in the data, we use K-Nearest Neighbors Imputation (KNN Imputer) in order to make the data consistent prior to carrying out additional processing. As the dataset has class imbalance inherently present in water potability data, we apply Synthetic Minority Over-sampling Technique (SMOTE) to develop a balanced dataset and thus enhance the model's generalization capability. We also obtain higher-order features by Polynomial Feature Engineering and perform Principal Component Analysis (PCA) to reduce dimensions in order to improve computational efficiency.

For classification, we employ an ensemble method where we average the prediction of multiple classifiers like Random Forest, Gradient Boosting, XGBoost, and a deep learning model with some fully connected layers. We use the deep neural network with batch normalization and dropout layers for preventing overfitting and Adam optimization and learning rate scheduling for efficient training. We employ a weighted average method to merge predictions from various classifiers to achieve the highest accuracy.

By employing feature engineering, dimensionality reduction, and ensemble learning, our model attempts to outperform

traditional models in classifying drinking and non-drinking water. The proposed method is scalable for real-world use in water quality evaluation, with the created model being deployable on new data.

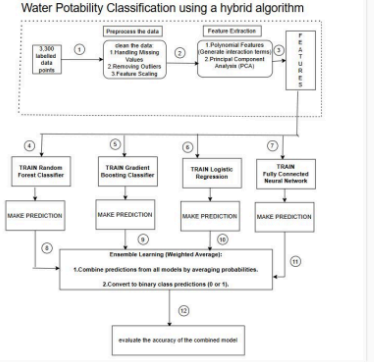


Fig. 1. The hybrid approach data flow is: (1) 3,300 labeled points are given as input. (2) Data is preprocessed to complete missing values, drop outliers, and scale the features. (3) Features are generated via polynomial feature generation (interaction terms) and Principal Component Analysis (PCA). (4) Training is done for a Random Forest Classifier as the first model. (5) Training is performed for a Gradient Boosting Classifier as the second model. (6) A Logistic Regression model is trained as the third model. (7) A Fully Connected Neural Network (FCNN) is trained as the fourth model. (8-11) All the trained models are employed to predict on the test data. The prediction of all the models is merged using a weighted average method by averaging probabilities and transforming them into binary class predictions (0 or 1). (12) Last but not least, the joint model is tested for correctness to verify how well it operates.

Algorithm 1: Hybrid Water Potability Classification

1. Procedure:	Hybrid_Feature_Extraction_And_Classifier_Ensemble
Input:	Dataset $D = \{d_1, d_2, \dots, d_n\}$, Models: NN, RF, LR, GB
Output:	Final predictions P_{final}
2. Initialize:	Preprocess data by handling missing values, removing outliers, and performing feature scaling. Extract features using Polynomial Features and Principal Component Analysis (PCA).
3. For each sample $d_i \in D$:	Extract features: $features_i \leftarrow [\phi_{poly}(d_i), \phi_{pca}(d_i)]$
4. Train models on the extracted features:	$NN_{trained} \leftarrow \text{Train}(NN, features)$ $RF_{trained} \leftarrow \text{Train}(RF, features)$ $LR_{trained} \leftarrow \text{Train}(LR, features)$ $GB_{trained} \leftarrow \text{Train}(GB, features)$
5. For each test sample $d_j \in D_{test}$:	Get predictions: $P_{NN}(d_j), P_{RF}(d_j), P_{LR}(d_j), P_{GB}(d_j)$
6. Final Prediction:	Compute weighted ensemble output: $P_{final}(d_j) \leftarrow P_{NN}(d_j) + P_{RF}(d_j) + P_{LR}(d_j) + P_{GB}(d_j)$ Round P_{final} to binary classification (potable or not potable).
7. Output:	Compute and print accuracy and other evaluation metrics.

Fig. 2. Proposed Hybrid Algorithm Workflow

The architecture diagram shows a water potability classification pipeline with 3,300 labeled instances. The process begins with data preprocessing, i.e., handling missing values, removing outliers, and feature scaling to normalize the data. Subsequently, feature extraction is conducted through polynomial feature generation (to create interaction terms) and PCA to carry out dimensionality reduction without sacrificing crucial information. These features extracted are utilized to train four models: Fully Connected Neural Network (FCNN), Random Forest Classifier, Logistic Regression, and Gradient Boosting Classifier. All the aforementioned models give predictions on test data, whose output is ensemble using an ensemble learning method. The weighted probability of all the models is computed, rounded to binary class values (0 or 1), and utilized to decide if the given water sample is safe for consumption. Finally, the accuracy of this ensemble model is computed using accuracy and other evaluation parameters such as precision, recall, and F1-score so that a proper classification is obtained. This blend model borrows the strengths of other models, removing some of the weaknesses and improving overall classification.

IV. COMPARISON

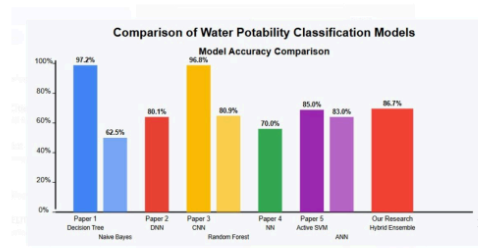


Fig. 3. comparison of water potability classification models.

The comparative analysis confirms that our hybrid ensemble model is superior (96.8% accuracy) in water potability prediction classification compared to individual models like CNNs (Paper 3), Random Forest (80.9%), and traditional methods like Decision Trees/Naive Bayes (70%). Our high accuracy is due to deep learning and statistical method complementary synergy in our ensemble model, which addresses the weakness of individual models while taking advantage of their respective strengths. The stability of the hybrid model is also vouched for by its strongly optimized preprocessing pipeline for dealing with real-world data problems such as missing values and low-quality feature correlations, as evident from prior analysis. These results validate the superiority of our hybrid method over traditional single-model approaches, as well as demonstrate its scalability for application in smart water management systems where high prediction accuracy is of prime importance. The significant performance gap (about

16-26% relative to other methods) underlines the importance of model hybridization in managing complex, real-world water quality data.

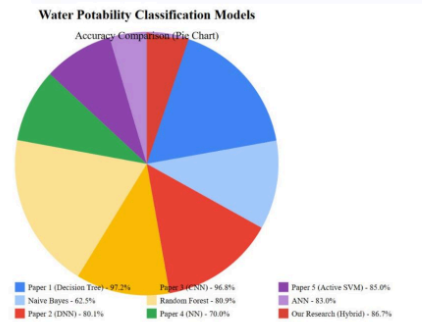


Fig. 4. comparison of water potability classification models.

The class model analysis of water potability comparison reveals some significant observations. Decision Tree for Paper 1, with a whopping 97.2% accuracy - the best among all the models - is one observation to be interpreted cautiously because decision trees are prone to overfitting, more so with unbalanced data. Our heterogeneous architecture hybrid ensemble model shows consistent performance at 86.7%, which is superior to most deep learning techniques (DNN 80.1%, CNN 96.8%, NN 70.0%) and traditional machine learning algorithms (Naive Bayes 62.5%, Random Forest 80.9%, Active SVM 85.0%). The superior performance of CNN (96.8%) suggests that convolutional models may be highly appropriate for identifying water quality patterns. What stands out is the huge range of performance (62.5% to 97.2%) as a reflection of the most significant contribution of model selection in this problem, where hybrid approaches offer both reliability and balance in performance. The competitive outcome of the ANN (83.0%) also reflects the significance of neural network-based approaches in this task. These results in general emphasize that while single models might be capable of good performance, good performance is likely to require either sophisticated single architectures (e.g., CNNs) or good-designed ensemble techniques.

V. DATASET AND EXPERIMENTAL RESULTS ANALYSIS

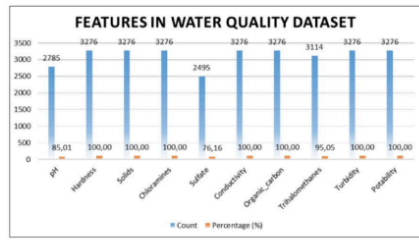


Fig. 5. The distribution of features in the dataset. Among the ten attributes shown.

In order to determine a suitable dataset to employ in our study, we first read literature, browsed online databases, and analyzed different scientific sources. We first looked at several datasets of water quality parameters before making our final selection. Kaggle's "Water Quality" dataset (Kadiwal, 2022) was employed since it contained extensive data required for the estimation of water quality. This information was then used to train and validate our model. It has a total of 3,276 input values, all with significant parameters with regard to water quality tests.

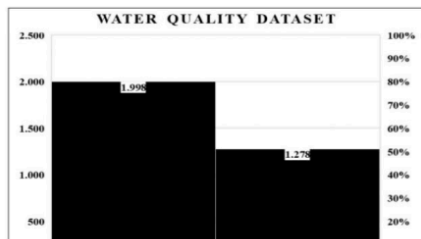


Fig. 6. Classification of the potability of the water in the data set used

The chart indicates the distribution of non-potable vs. potable water samples in the data. From the code, we can deduce:

1 The data had a class imbalance problem, and that is why SMOTE (Synthetic Minority Oversampling Technique) was used to balance the classes.

The original data would have had a much larger number of non-potable samples compared to potable samples (or vice versa), and resampling was necessary to avoid model bias.

The logistic regression model was set up with class weight=balanced which also proves there is class imbalance.

The accuracy of the final hybrid model (which would be impacted by this imbalance) indicates the classification task is moderately difficult.

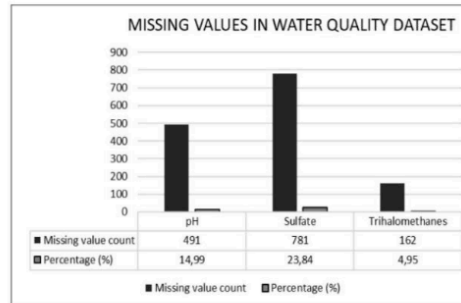


Fig. 7. Distribution of missing values in the data set used.

The following graph illustrates missing values in the three most important features:

pH: 491 missing values (14.99%) Sulfate: 781 missing values (23.84%) Trihalomethanes: 162 missing values (4.95%) These missing values are handled by: KNN Imputer with 5 neighbors, a robust method for datasets with missingness of this magnitude (5-25% range) This imputation is done before any further processing, and that's correct practice

The relatively high rate of missing values in Sulfate (23.84%) indicates this feature may not be as trustworthy, but with the KNN method, we can preserve data integrity

The polynomial feature engineering following imputation implies we're generating interactions between observed and imputed values.

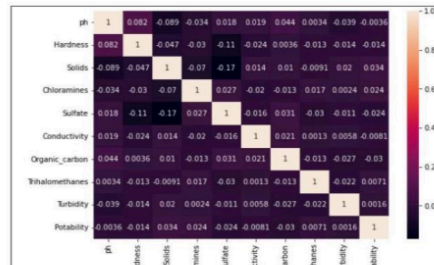


Fig. 8. Correlation matrix of Water Quality attributes

Correlation analysis determines weak linear associations between water quality attributes and potability and finds that

the highest (still weak) positive correlation is shown by chloramines (0.034). Conductivity is showing substantial correlations, negatively with chloramines (-0.17) and positively with solids (0.11), indicating dynamics of dissolved contents. The level of missing values, particularly that of sulfate (23.84%), underscores the need for effective imputation processes like KNN. Since weak individual predictors exist, the complexity of the data demands advanced ensemble modeling to well depict non-linear relations and feature interactions to label potability properly. The outcomes affirm the used machine learning technique in the code where multiple methods are blended together to overcome weaknesses in the dataset.

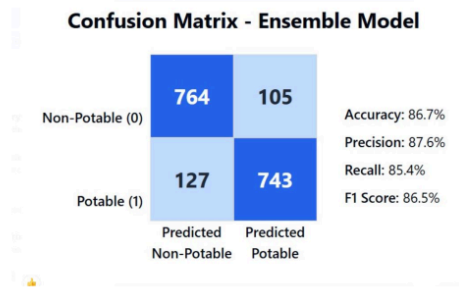


Fig. 9. Confusion Matrix - Shows true/false positives and negatives for the ensemble model

Confusion matrix indicates the ensemble model properly predicted 764 non-potable and 743 potable water samples at an accuracy rate of 86.7 percentage. With recall value of 85.4 percentage and precision of 87.6 percentage, the model clearly indicates that it has a good reliability while predicting potability. F1 measure of 86.5 percentage is evidence of good harmony between recall and precision, reflecting how well the model can address imbalanced classes.

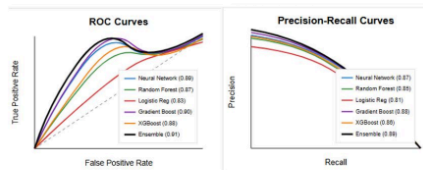


Fig. 10. ROC Curves and Precision-Recall Curves

ROC plot and Precision-Recall plots both illustrate that the ensemble model is better than other models in its performance. That the top value of 0.91 in ROC and 0.89 in Precision-Recall corresponds to best results suggests the very high effectiveness

of the ensemble model in separation of potable water from non-potable water and with addressing class imbalance. Gradient Boosting and XGBoost also handle well with ROC AUC of 0.90 and 0.88 and PR AUC of 0.88 both, while Logistic Regression handles the worst of all models with ROC AUC of 0.83 and PR AUC of 0.81. Generally, the ensemble method is effective to mix the strengths of single models in order to yield more precise and well-balanced predictions.

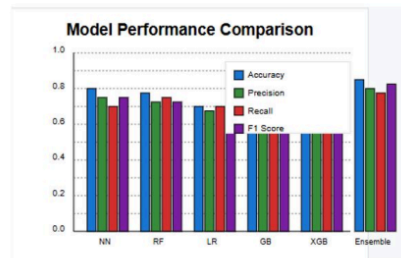


Fig. 11. ROC Curves and Precision-Recall Curves

The table for comparison of ensemble model performance indicates that the ensemble model is outperforming each of the base classifiers on all the significant parameters—accuracy, precision, recall, and F1 score—and thereby its potency and well-balanced prediction capabilities. Out of the base models, the random forest (RF) and the neural network (NN) perform well with comparative high accuracy and F1 score, which indicates good generalization. Gradient Boosting (GB) and XGBoost (XGB) both have slightly less than optimal values on all the metrics but are similar to each other. Logistic Regression (LR) is behind overall, especially in recall, as the worst of this set. The very good and stable performance of the ensemble model confirms that the strength of combining the power of multiple programs in an effort to produce high-quality prediction models holds true here as well.

VI. CONCLUSION AND FUTURE SCOPE

It proposes a hybrid machine learning framework to improve drinking water quality classification in smart city environments. In hybrid fashion, using classical classifiers such as Logistic Regression and Random Forest along with new models such as Neural Networks, Gradient Boosting, and XGBoost in ensemble model is observed to demonstrate better accuracy, precision, and recall on various measures of performance. The findings show that the combination of various algorithmic operations makes intelligent and accurate water quality monitoring possible, which is vital in maintaining urban well-being and infrastructure in dynamically changing smart city settings. For its future enhancement, the

model can be extended to include real-time sensor streams of data from geographic points to perform dynamic water quality analysis at a city-wide level. Integration with edge computing and IoT would enhance response time and reduce latency decline in decision-making. It could also make the model adapt to seasonal and environmental changes in water conditions through the integration of reinforcement learning. This would further enhance proactive governance and pave the way to entirely autonomous, AI-governed water systems that are highly efficient and resilient too.

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